

# Infron 与 OpenRouter 路由、缓存、成本与流式性能 A/B 实验报告

## 摘要

本报告评估 `deepseek/deepseek-v4-pro` 在 Infron 与 OpenRouter 的 Prompt Caching 工作负载表现。实验使用 `/v1/chat/completions`，保留模型与平台默认 reasoning/thinking 行为，并启用 prompt 长度分层。

过滤后保留 791 个配对样本 和 3164 条请求级观测，剔除 18 条记录。

核心结论：Infron 在全部路由模式下实际成本更低，并在端到端 E2E 时延上领先；OpenRouter 在全部路由模式下吞吐量和 Streaming TTFT 更优。缓存命中率在不同路由模式下接近或互有领先。

## 质量门禁

| 项目                   | 值                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------|
| 模型                   | <code>deepseek/deepseek-v4-pro</code>                                                               |
| API 协议               | 仅 <code>/v1/chat/completions</code>                                                                 |
| 组数 / 轮数              | 4 组 x 50 轮                                                                                          |
| Prompt 分层            | <code>short</code> about 1500, <code>medium</code> about 8000, <code>long</code> about 32000 tokens |
| Reasoning / thinking | 平台与模型默认行为；请求中不显式禁用                                                                                  |
| 配对规则                 | strict sort/group/round pair, first/second <code>usage.prompt_tokens</code> 差异均不超过 50               |
| 保留配对样本               | 791                                                                                                 |
| 请求级观测                | 3164                                                                                                |
| 剔除记录                 | 18                                                                                                  |

## 路由级结果矩阵

| 路由模式 | 配对样本 | Token 命中率                         | 实际成本                                      | 吞吐量                                          | E2E 时延                                     | TTFT                                      |
|------|------|-----------------------------------|-------------------------------------------|----------------------------------------------|--------------------------------------------|-------------------------------------------|
| 吞吐优先 | 200  | Infron 90.52% / OpenRouter 95.98% | Infron \$1.205367 / OpenRouter \$2.110727 | Infron 2.805 tok/s / OpenRouter 39.962 tok/s | Infron 4557.65 ms / OpenRouter 10533.90 ms | Infron 4139.22 ms / OpenRouter 3138.79 ms |

| 路由模式       | 配对样本 | Token 命中率                         | 实际成本                                      | 吞吐量                                          | E2E 时延                                     | TTFT                                      |
|------------|------|-----------------------------------|-------------------------------------------|----------------------------------------------|--------------------------------------------|-------------------------------------------|
| 价格优先       | 198  | Infron 99.44% / OpenRouter 98.26% | Infron \$0.337858 / OpenRouter \$1.693868 | Infron 3.197 tok/s / OpenRouter 38.359 tok/s | Infron 4023.15 ms / OpenRouter 10514.30 ms | Infron 3350.61 ms / OpenRouter 3210.02 ms |
| 端到端时延优先    | 196  | Infron 91.27% / OpenRouter 92.33% | Infron \$1.109211 / OpenRouter \$1.780845 | Infron 2.949 tok/s / OpenRouter 39.703 tok/s | Infron 4305.92 ms / OpenRouter 10042.08 ms | Infron 3863.79 ms / OpenRouter 3191.40 ms |
| 流式 TTFT 优先 | 197  | Infron 98.59% / OpenRouter 99.45% | Infron \$0.589968 / OpenRouter \$1.442047 | Infron 2.790 tok/s / OpenRouter 36.153 tok/s | Infron 4588.39 ms / OpenRouter 10079.76 ms | Infron 4099.19 ms / OpenRouter 3196.65 ms |

API 协议记录

| API 协议           | Endpoint             | 说明                                                             |
|------------------|----------------------|----------------------------------------------------------------|
| chat_completions | /v1/chat/completions | 本轮仅保留该标准 Chat Completions 协议；不包含 /v1/messages 或 /v1/responses。 |

Prompt 长度分层缓存表现

| 分层     | 配对样本 | Token 命中率                       | 调用命中率                           | E2E 时延                              | TTFT                                | Reasoning tokens        |
|--------|------|---------------------------------|---------------------------------|-------------------------------------|-------------------------------------|-------------------------|
| short  |      | Infron 0.00% / OpenRouter 0.00% | Infron 0.00% / OpenRouter 0.00% | Infron 0.00 ms / OpenRouter 0.00 ms | Infron 0.00 ms / OpenRouter 0.00 ms | Infron 0 / OpenRouter 0 |
| medium |      | Infron 0.00% / OpenRouter 0.00% | Infron 0.00% / OpenRouter 0.00% | Infron 0.00 ms / OpenRouter 0.00 ms | Infron 0.00 ms / OpenRouter 0.00 ms | Infron 0 / OpenRouter 0 |
| long   |      | Infron 0.00% / OpenRouter 0.00% | Infron 0.00% / OpenRouter 0.00% | Infron 0.00 ms / OpenRouter 0.00 ms | Infron 0.00 ms / OpenRouter 0.00 ms | Infron 0 / OpenRouter 0 |

Reasoning / Thinking 观测

本轮不显式禁用或强制 reasoning/thinking；reasoning tokens 按响应 telemetry 观测。

| 路由模式       | Infron reasoning tokens | OpenRouter reasoning tokens | Infron 平均/请求 | OpenRouter 平均/请求 |
|------------|-------------------------|-----------------------------|--------------|------------------|
| 吞吐优先       | 0                       | 162216                      | 0.0000       | 405.5400         |
| 价格优先       | 0                       | 153691                      | 0.0000       | 388.1086         |
| 端到端时延优先    | 0                       | 150585                      | 0.0000       | 384.1454         |
| 流式 TTFT 优先 | 0                       | 138211                      | 0.0000       | 350.7893         |

# 可复现性引用

| 工件             | GitHub 路径                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SHA256 / 说明                                                |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Summary        | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/summary.json">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/summary.json</a>                                                   | 0984289dd19dcfd12644cbe0d4e2473b57ee98cf65ced71209d32710e8 |
| 配对数据集 CSV      | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_pairs.csv">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_pairs.csv</a>                                     | 858cdad534209908d7dbce59a92205040a3a008e639bf860d527b4358b |
| 请求级数据集 JSONL   | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_requests.jsonl">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_requests.jsonl</a>                           | c1958ba57cc831fad6677f10937e6ce984137f351f2e7b1f1f05e34558 |
| 剔除记录审计         | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/records_excluded.json">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/records_excluded.json</a>                                 | 104a6a0249e8026ea16e841ebdff3e855e6a526297db4a5ef6441f3999 |
| Benchmark 执行源码 | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/rerun_routing_sort_cache_cost_ab.py">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/rerun_routing_sort_cache_cost_ab.py</a>     | a485c5116e99375e6b171d0328e5a423babd1ee6b206c72f54053a3a04 |
| HTML 报告渲染源码    | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/render_glm52_deepseek_style_report.py">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/render_glm52_deepseek_style_report.py</a> | e5bb0d32361d70ee688e0d0a0da3302a1e542e33e53876c905814eb5c2 |

| 工件         | GitHub 路径                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SHA256 / 说明                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 测试源码       | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/test_rerun_routing_sort_cache_cost_ab.py">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/test_rerun_routing_sort_cache_cost_ab.py</a> | c086ddc5d0a9a91eba82b7e8767d7bddf2f1ca4a28af0a87056b702882 |
| Reports 目录 | <a href="https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports">https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports</a>                                                                             | 中英文 HTML / Markdown / PDF 报告                               |
| Data 目录    | <a href="https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data">https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data</a>                                                                                   | 配对数据、请求级 telemetry、summary 与剔除样本                           |
| Manifest   | <a href="https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/metadata/manifest.json">https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/metadata/manifest.json</a>                                               | 文件大小与 SHA-256 checksum                                     |

数据集引用： `business_representative` 内置代表性业务模板；请求级导出见 `benchmark_requests.jsonl`。

在线 HTML：中文 [https://infronai.github.io/prompt-cache-bench/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study\\_deepseek-v4-pro\\_infron-vs-openrouter\\_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions\\_2026-07-04.zh.html](https://infronai.github.io/prompt-cache-bench/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_deepseek-v4-pro_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.zh.html)；英文 [https://infronai.github.io/prompt-cache-bench/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study\\_deepseek-v4-pro\\_infron-vs-openrouter\\_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions\\_2026-07-04.en.html](https://infronai.github.io/prompt-cache-bench/experiments/deepseek/deepseek-v4-pro/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_deepseek-v4-pro_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.en.html)。